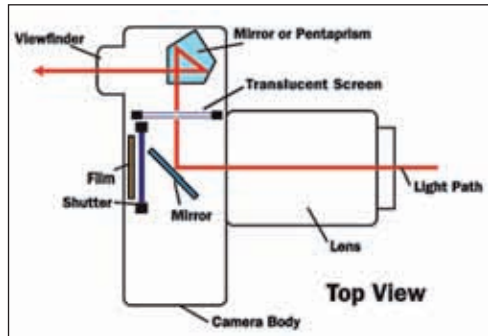


prism (or penta-mirror). When we see through the viewfinder, we see the light reflected by this penta-prism, which focuses it, so as to be directly observable via the photographer's eye. As you press the shutter button, the viewfinder is blocked for an instant (called viewfinder blackout). This happens because the inclined mirror reflecting the light swings upwards, allowing the light that comes in through the lens to hit the sensor directly thereby exposing it and capturing the exact image.



Credits: HowStuffWorks

While this process may sound a tad too long, any entry-level SLR will be able to achieve the entire operation three times a second or more.

This simple yet very effective accuracy in composing pictures from the operative difference between SLRs and point and shoots is actually responsible for their preference by most serious photographers or even photography enthusiasts, resulting in an ecosystem of lenses and accessories around them. Infact, interchangeable lenses are one of the biggest draws towards using an SLR. From even advanced point and shoot cameras, it's often impossible to get such superior results as can be had thanks to the ability of SLRs to employ an array of custom lenses which serve specific needs instead of a single, general purpose.

While point and shoots use contrast-detection technology for auto-focusing, SLRs often employ phase-detection technology – a much faster method. Though this allows for less “focus searching”, it needs a special sensor in the optical path which, though present in SLRs, is missing from point and shoots. P&S cameras are slower in some implementations because they instead have a main sensor that creates a preview on the electronic viewfinder using the contrast-detect method.

Not just in lenses and sensors, some of the biggest differences of the two cameras lie in their build types. Despite the complex optics and more elaborate internal designs, SLRs are typically the more rugged and modular